Technical Report Writing

on

1st Order and 1st Degree differential equations

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📚 **ABSTRACT:** This report explores techniques to solve first-order and first-degree differential equations, including exact, linear, and nonlinear forms. Various mathematical methods are applied to solve given problems, highlighting their importance in engineering modeling and real-world applications.

🧩 **INTRODUCTION:** Differential equations play a crucial role in engineering, physics, and applied sciences as they describe relationships involving rates of change. First-order and first-degree differential equations are among the simplest yet most powerful tools for modeling real-world phenomena such as population growth, electrical circuits, and fluid flow.

This report focuses on solving different types of first-order differential equations using systematic methods such as exact equations, integrating factors, and substitutions. The aim is to develop a clear understanding of solution techniques and their practical relevance. The report is structured to include theory, step-by-step solutions, and analytical discussion of each problem.

📌 **PROCEDURE AND DISCUSSION:**

1. **<u>Exact Differential Equation</u>**

Given: $(2x \log x - xy)\, dy + 2y\, dx = 0$

🔎 **Method:**

Compare with: $M(x,y)dx + N(x,y)dy = 0$

Check exactness: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

🎲 **Solution Idea:**

- If exact → find function F(x,y)
- Integrate M w.r.t x and N w.r.t y

♟️ **Final Result:** $F(x,y) = C$

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2. **<u>Homogeneous Differential Equation</u>**

Given: $(x^2y)dx - (x^3+y^3)dy = 0$

🔎 **Method:**

Use substitution: $v = \frac{y}{x}$

$\Rightarrow y = vx$

🎲 **Process:**

- Convert into separable form
- Integrate both sides

♟️ **Final Result:** General solution in implicit form

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3. **<u>Nonlinear Equation Transformation</u>**

   Given: $y(1 + xy)dx + x(1 - xy)dy = 0$

🔎 **Method:**

- Try substitution $xy = t$
- Reduce to simpler differential form

♟️ **Outcome:**

- Simplified integration leads to implicit solution

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4. **<u>Linear Differential Equation</u>**

   Given:

$$\text{Cot}^3x\frac{dy}{dx} - 3y = \cos^3x + \sin^3x$$

🎲 **Standard Form:**

$$\frac{dy}{dx} + P(x)y = Q(x)$$

🔎 **Method:**

- Find Integrating Factor (IF):

$$\text{IF} = e^{\int P(x)dx}$$

♟️ **Solution:**

$$y \cdot \text{IF} = \int Q(x) \cdot \text{IF}\, dx + C$$

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5. **<u>Bernoulli's Equation</u>**

   Given: $\frac{dy}{dx} - y = y^2(\sin x + \cos x)$

🎲 **Form:** $\frac{dy}{dx} + Py = Qy^n$

🔎 **Method:**

- Divide by y^2
- Substitute: $z = \frac{1}{y}$

♟️ **Convert → Linear equation**

Solve using integrating factor.

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6. **<u>Logarithmic Transformation Equation</u>**

Given: $\frac{dz}{dx} + (\frac{z}{x}) \log z = (\frac{z}{x}) (\log z)^2$

🔎 **Method:**

Substitute: $\log z = t$

♟️ **Simplification:**

- Convert to separable form
- Integrate both sides

🔬 **CONCLUSION**

This report demonstrates various techniques to solve first-order differential equations, including exact, homogeneous, linear, and Bernoulli forms. These methods are essential for solving real-world engineering problems. A strong understanding of these approaches enhances analytical thinking and mathematical modeling skills.

📚 **REFERENCES**

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